

EXERCISE SHEET 7

GROUP AND RING THEORY

MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

7.1. Consider the diagram of R -modules and R -homomorphisms

$$\begin{array}{ccc} & F & \\ \swarrow h & \downarrow f & \\ M & \xrightarrow{g} N & \longrightarrow 0 \end{array}$$

where the map g is surjective. Suppose F is free. Show that there exists an R -homomorphism $h : F \rightarrow M$ such that $gh = f$. (Here gh is the composition of the two maps.)

7.2. Consider the ring $R = \mathbb{Z} \times \mathbb{Z}$. Is the R -module R free? Is the R -submodule $\mathbb{Z} \times \{0\}$ also free?

7.3. Let R be a ring with unity. An R -module P is *projective* if there is another R -module Q such that the direct sum of P and Q is a free R -module. Are all free R -modules projective? Is the converse true?

7.4. Let R be a commutative ring with unity, and let $e \neq 0, 1$ be an idempotent in R . Show that Re is a projective R -module. Is it a free R -module?

EXTRA QUESTIONS

7.5. An onto homomorphism $g : A \rightarrow B$ from an R -module A to an R -module B is said to *split* if $\ker g$ is a direct summand of A (i.e. $A = \ker g \oplus C$ for some submodule C of A). Show that if B is free and $g : A \rightarrow B$ is an onto homomorphism, then g splits.

7.6. (i) Let A be a ring with unity. If K is any left ideal of a ring $R = M_n(A)$, the $n \times n$ matrix ring over A , show that $E_{11}K$ is a submodule of the free A -module $E_{11}R$.

(ii) If L is any A -submodule of $E_{11}R$, show that $\sum_i E_{i1}L$ is a left ideal of R .

(iii) Verify that $K \longleftrightarrow L$ is a 1 - 1 correspondence that preserves inclusions.